Islamic University of Technology (IUT)

Electrical and Electronic Engineering Department

COMPLEX ENGINEERING PROBLEM

Course No.: **EEE 4401** **Course Title:** Power System I

Academic Year: 2022- 2023 **Semester:** Summer

Name(s) of the Course Instructor(s): Muhammad, Assistant Professor, EEE Department.

1. Statement of Complex Engineering Problem. [**Marks: 15** **COs: CO3**

Inclusion: Assignment]

Part A: Select Suitable Power Systems for Load Flow Analysis

Design a power system for Load Flow Analysis by conducting literature review of relevant research papers and books. The power system must have the following specifications:
a) There must be a minimum of **5 bus bars** in the power system.
b) There must be **at least two** generation buses in the system.
c) Total power demand in buses needs to be satisfied as **P = [200 250]** MW and **Q = [80 100]**
d) Solution must be converged and total power **loss must be minimized**.

Part B: Load Flow Analysis

Task 1: Load Flow Analysis using MATLAB

Solve the designed/selected system with reference to the following:

a) Prepare input files in standard format for line data for MATLAB.

b) Develop admittance matrix, $\mathbf{Y_{bus}}$ for the proposed system both in rectangular and polar form.

c) Perform complete load flow analysis correct up to three decimal places using MATLAB codes available in relevant text books and prepare comparison table showing the results. Evaluate results and write your comments about it.

Task 2: Load Flow Analysis using Power World Simulator

a) Depict the real and reactive power flows and losses using the software and write your comments about it.
b) Prepare comparison table showing the results.

Solution of the Complex Engineering Problem.

List Program Outcomes (POs) as well as their attributes that have been addressed in the Complex Engineering Problem.

	Statement	Put Tick (√)	Different Aspects	Put Tick (√)
PO1	**Engineering knowledge:** Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex electrical and electronic engineering problems.			
PO2	**Problem analysis:** Identify, formulate, research literature and analyse complex electrical and electronic engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.	√		
PO3	**Design/development of solutions**: Design solutions for complex electrical and electronic engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.		Public health	
			Safety	
			Cultural	
			Societal	
			Environmental	
PO4	**Investigation:** Conduct investigations of complex electrical and electronic engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	√	Design of experiments	
			Analysis and interpretation of data	√
			Synthesis of information	
PO5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex electrical and electronic engineering problems, with an understanding of the limitations.	√		
PO6	**The engineer and society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex electrical and electronic engineering problems.		Societal	
			Health	
			Safety	
			Legal	
			Cultural	
PO7	**Environment and sustainability:** Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex electrical and electronic engineering problems in societal and environmental contexts.		Societal	
			Environmental	
PO8	**Ethics:** Apply ethical principles embedded with religious values, professional ethics and responsibilities, and norms of electrical and electronic engineering practice.		Religious values	
			Professional ethics and responsibilities	
			Norms	

PO9	**Individual work and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.		Diverse teams	
			Multi-disciplinary settings	
PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.		Comprehend and write effective reports	
			Design documentation	
			Make effective presentations	
			Give and receive clear instructions	
PO11	**Project management and finance:** Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.		Engineering management principles	
			Economic decision-making	
			Manage projects	
			Multidisciplinary environments	
PO12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the context of electrical and electronic engineering related technological change.			

Attributes of Knowledge Profile (K1 – K8) that have been addressed in the Complex Engineering Problem.

K	Knowledge Profile (Attribute)	Put Tick (√)
K1	A systematic, theory-based understanding of the natural sciences applicable to the discipline	
K2	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline	
K3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline	√
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline	
K5	Knowledge that supports engineering design in a practice area	√
K6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline	
K7	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability	
K8	Engagement with selected knowledge in the research literature of the discipline	

Attributes of ranges (P1 – P7) that have been addressed in the Complex Engineering Problem.

P	Range of Complex Engineering Problem Solving	Put Tick (√)
Attribute	Complex Engineering Problems have characteristic P1 and some or all of P2 to P7:	
Depth of knowledge required	**P1:** Cannot be resolved without in-depth engineering knowledge at the level of one **or** more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach	√
Range of conflicting requirements	**P2:** Involve wide-ranging or **conflicting technical**, engineering and other issues	√
Depth of analysis required	**P3:** Have **no obvious solution** and require abstract thinking, originality in analysis to formulate suitable models	√
Familiarity of issues	**P4:** Involve infrequently encountered issues	
Extent of applicable codes	**P5:** Are outside problems encompassed by standards and codes of practice for professional engineering	
Extent of stakeholder involvement and conflicting requirements	**P6:** Involve diverse groups of stakeholders with widely varying needs	
Interdependence	**P7:** Are high level problems including many component parts or sub-problems	

Attributes of ranges of Complex Engineering Activities (A1 – A5) that have been addressed in the Complex Engineering Problem.

A	Range of Complex Engineering Activities	Put Tick (√)
Attribute	Complex activities means (engineering) activities or projects that have some or all of the following characteristics:	
Range of resources	**A1:** Involve the use of diverse resources (and for this purpose resources include people, money, equipment, **materials, information and technologies**)	√
Level of interaction	**A2:** Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues	
Innovation	**A3:** Involve creative use of engineering principles and research-based knowledge in novel ways	
Consequences for society and the environment	**A4:** Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation	
Familiarity	**A5:** Can extend beyond previous experiences by applying principles-based approaches	
Complex Engineering Activities have not been addressed		